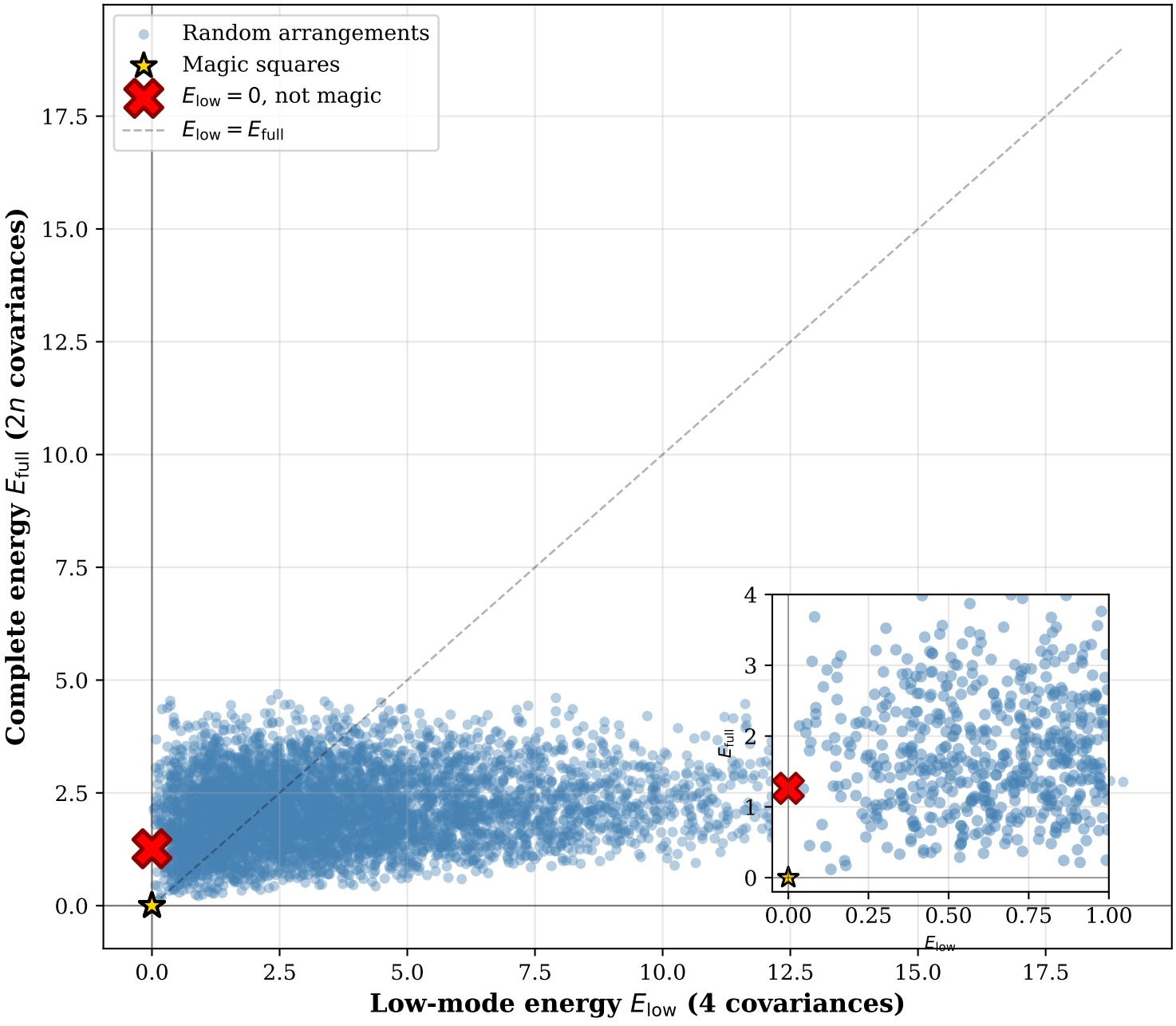
Comparing Low-Mode and Complete Energies ($n = 4$)
Random arrangements
Magic squares
$E_{\text{low}} = 0$, not magic
$E_{\text{low}} = E_{\text{full}}$
Complete energy E_{full} ($2n$ covariances)
Low-mode energy E_{low} (4 covariances)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
E_{full}
E_{low}
0
1
2
3
4
0.00
0.25
0.50
0.75
1.00